

AHMED MARWAN ABDULHABEB AL-QERSHI

OPERATIONS RESEARCH ANALYST · STOCHASTIC OPTIMIZATION & SCIENTIFIC SOFTWARE

RESEARCH PROFILE

My work sits between operations research and scientific software. I have an Industrial Engineering background, a thesis-based M.Sc. in Electrical and Computer Engineering, and professional experience implementing stochastic optimization methods in GAMS Py. I am especially interested in decomposition methods and multiscale energy-system models.

RESEARCH INTERESTS

- Multistage stochastic programming
- SDDP and Benders decomposition
- Multiscale energy-system modelling
- Distribution grids under uncertainty
- Scientific software for optimization

CORE METHODS

- LP, MILP and NLP modelling
- Stochastic decomposition
- Discrete-event simulation
- Simulation-optimization
- Mathematical and economic modelling

LANGUAGE SKILLS

- Arabic – Native
- English – Fluent
- Turkish – Good

ACADEMIC DISTINCTIONS

- First-ranked Industrial Engineering graduate, 2022
- High Honour graduate and Certificate of Merit
- AGU Top Achiever, 2020–2022

EDUCATION

M.SC. ELECTRICAL AND COMPUTER ENGINEERING 09.2023 – 05.2025
Abdullah Gül University, Kayseri, Türkiye · 120 ECTS · GPA 3.96/4.00

- Two-year, thesis-based second-cycle degree, taught and examined entirely in English.
- **Thesis (45 ECTS):** *Neural Insights into Drug Repositioning: A Literature-Based Framework Using Word Embeddings and Siamese Networks*. Sole author; supervisor: Asst. Prof. Mehmet Gökhan Bakal.
- Designed the TensorFlow/Keras research pipeline and evaluated 576 experimental configurations, reaching 87.66% validation accuracy and 83.2% test accuracy.

B.SC. INDUSTRIAL ENGINEERING 08.2018 – 06.2022
Abdullah Gül University, Kayseri, Türkiye · 245 ECTS · GPA 3.84/4.00

- Graduated with High Honour and ranked first among the department's 2022 graduates; degree taught and examined entirely in English.
- **57 ECTS in operations research and optimization:** Stochastic Models (A), Deterministic Optimization (A), Mathematical Modeling (A–), System Simulation (A), Decision and Risk Analysis (A), and related analytics and production courses.
- Further preparation: 34 ECTS in mathematics, probability and statistics; 23 ECTS in computing; 32 ECTS of supervised industry work, including a 20-ECTS capstone.

SELECTED RESEARCH & SCIENTIFIC SOFTWARE

SDDP FRAMEWORK FOR GAMS PY 02.2026 – 07.2026
Core developer and primary implementer · GAMS Software GmbH · ongoing maintenance

- Researched Stochastic Dual Dynamic Programming, first reproducing a multistage hydrothermal power-planning example based on Vattenfall Energy Trading data and then generalizing it into a reusable GAMS Py framework.
- Implemented sampled forward passes, backward scenario solves, LP-dual-based value-function cuts, lower-bound tracking, convergence checks and the interface for user-defined stages, states and discrete uncertainty.
- When the first general version proved too slow, profiling located the bottleneck in repeated Python–GAMS round trips. I added GUSS (Gather–Update–Solve–Scatter) support so related scenario solves could be batched.
- Wrote tests, examples and documentation. The reviewed implementation was released in GAMS Py 1.25.0.

gamspy.readthedocs.io/en/latest/user/sddp/sddp.html

DYNAMIC CGE MODELLING PLATFORM 07.2025 – 05.2026
Project manager and lead developer · Kaizen Consulting

- Built a reusable dynamic computable general equilibrium platform covering formulation, calibration, solver integration, data workflows and the analyst-facing software.
- Validated it on a 10-year Saudi-economy case with 20 sectors and 84 commodities. Coordinated a six-person team of three senior economists and three developers.

TECHNICAL SKILLS

MODELLING SYSTEMS

GAMS · GAMSPy

PROGRAMMING

Python (primary) · Julia (working knowledge) · SQL · C# · Java

OPTIMIZATION SOLVERS

CPLEX · Gurobi · HiGHS · CONOPT

SIMULATION

Simio · OptQuest · discrete-event modelling

RESEARCH SOFTWARE

Git · unit testing · API design · profiling · documentation · Docker

MACHINE LEARNING

TensorFlow/Keras · PyTorch · scikit-learn · FastText

JULIA EXPERIENCE

Studied SDDP.jl's API and architecture when designing GAMSPy's SDDP interface; able to read and write Julia.

TEACHING

VOLUNTEER TEACHING ASSISTANT

Stochastic Models · Abdullah Gül University · 10.2021-01.2022

Joined practical sessions, worked through numerical examples and supported students outside class after earning an A in the course. Recorded public tutorials on Markov chains and Poisson processes.

REFERENCES

DR STEPHEN J. MAHER

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PROFESSIONAL EXPERIENCE

OPERATIONS RESEARCH ANALYST

01.2023 – PRESENT

GAMS Software GmbH · Remote

- Core developer on GAMSPy, GAMS's Python modelling library, with primary responsibility for the SDDP framework described on page 1.
- Enhanced GAMSPy's piecewise-linear functions API with support for additional curve types and a simpler interface; also contributed neural-network and ReLU formulations and bound propagation.
- Implemented most worked examples in the GAMSPy model library and fixed defects identified while translating established GAMS models.
- Additional work includes the Connect data-integration framework, API testing, documentation and solver-interface maintenance.

OPERATIONS RESEARCH ANALYST · ERASMUS TRAINEE

09.2022 – 12.2022

GAMS Software GmbH · Braunschweig, Germany

- Researched multi-objective optimization and implemented an initial executable example based on the sandwich algorithm, with technical presentations and documentation.

ECONOMIC MODELLING & SOFTWARE DEVELOPMENT CONSULTANT

07.2025 – PRESENT

Kaizen Consulting · Project-based, part-time · Remote

- Develop mathematical and economic models together with the software used to calibrate them, manage data, configure scenarios and inspect results.
- Projects include the dynamic CGE platform, a Tourism Satellite Account toolkit for a national Ministry of Tourism, and a structured tender-pricing platform.

SELECTED APPLIED OPERATIONS RESEARCH PROJECTS

HYBRID SIMULATION–OPTIMIZATION · STRYKER

06.2021 – 05.2022

- Plant-wide value-stream mapping identified the plastic-injection station as the main source of lead time. Built a Simio model for 90 parts, 72 molds and three machines and used OptQuest to tune a pull-based (Q,R) policy under a 90% service-level requirement.
- Integrated Simio with a Python/Gurobi production scheduler through C# and a database. Year-long 20-ECTS capstone, grade A.

PRODUCTION SCHEDULING · HES KABLO

11.2020 – 05.2021

- Used full-line simulation to identify three machines at approximately 98% utilization, then formulated the resulting flow shop as a makespan-minimizing MILP.
- Developed a sub-second heuristic extending Johnson's rule. Across 30 demand samples it matched the proven optimum 19 times; the mean deviation was 2.6 minutes. The deployed schedules corresponded to an estimated 32% increase in effective production capacity.

CUTTING-STOCK OPTIMIZATION · FEMAS METAL

02.2020 – 05.2020

- Led a three-person team; enumerated feasible patterns in Java and optimized their selection in GAMS, reducing trim loss by more than 80%.

PUBLICATIONS & RESEARCH OUTPUTS

Alqershi, A. and Bakal, G. "A Siamese Neural Network Framework for Computational Drug Repositioning Using Literature-Derived Biomedical Knowledge". Revised manuscript under review at the *Journal of Computational Science*. First author.

Alqershi, A., Usul, B., Bussieck, M., Soytürk, M. and Dirkse, S. (2025). "Speed vs. Guarantees: A Practical MIP–NLP Trade-off for NN Robustness". *GAMS Technical Blog*. Lead author. [Article](#)

GAMSPy SDDP framework (2026). Core developer and primary implementer; released in GAMSPy 1.25.0 with tests, examples and user documentation. [Documentation](#)